IS THERE A PUZZLE IN UNDERWATER MORTGAGE DEFAULT?

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BoC Housing workship | 2026 Sep 18

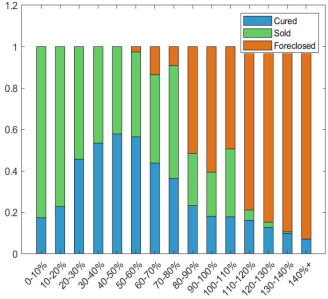
This Paper in Brief

- Puzzle: We rarely observe underwater mortgage default in the data
- To match models with the data, some models require a very high utility cost, worth 25 percent of lifetime consumption
- Instead, if we measure the value of delinquency/foreclosure better, we can match underwater mortgage delinquency with a much smaller utility cost
  - rent cost is stable; rent is relatively expensive when house prices fall

Should not measure house value vs mortgage value; instead measure $\underbrace{v(\text{def}) \text{ vs } v(\text{not def})}_{\text{rent, moving costs ...}}$

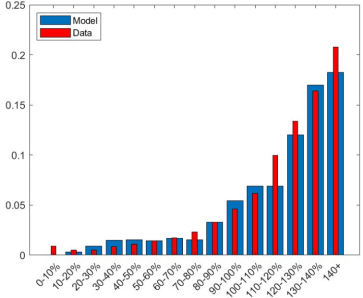
Still puzzling?

Most underwater borrowers who default on payments do foreclose



(c) Foreclosure/Sale/Cure Rates by LTV among Defaulted Loans

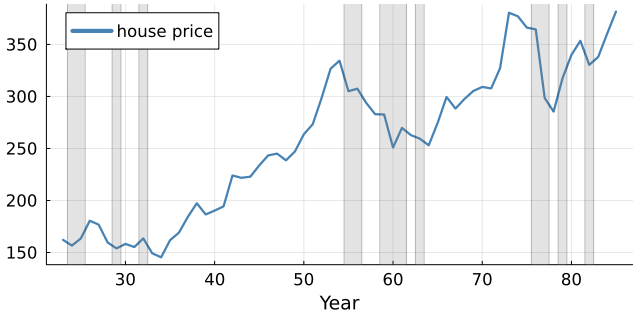
But why do only 10–20% of underwater borrowers default?



(a) Default Rate

- Is rent expensive enough to give up 40% of housing value?
- Measure other costs carefully: smaller rental housing, cannot buy a house for the next 7 years

House price expectations and frictions in housing transaction

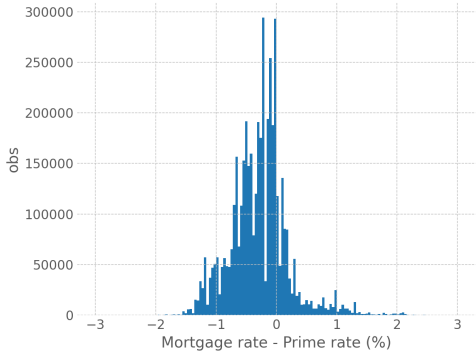


- House price expectations affect the value of foreclosure; if borrowers expect prices to stay low for an extended period while they are underwater, they have more incentive to foreclose
- While this is not a feature of this model, housing transactions and foreclosure take time, and there is uncertainty about future house prices

Quantitative implication of homogeneous mortgage rate

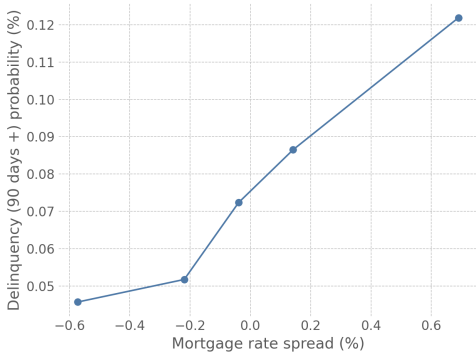
- In the model, all borrowers face the same mortgage rate

But mortgage rates are quite dispersed



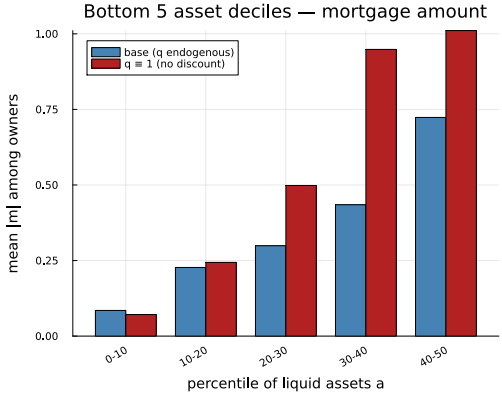
Data: OSFI

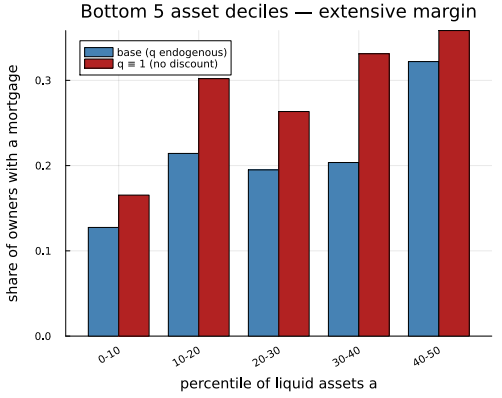
...and higher rates are associated with higher delinquency rates



Data: OSFI, TransUnion

Quantitative implication of homogeneous mortgage rate





'The Macroeconomic Effects of Debt Relief Policies during Recessions', Lee, 2025

- Without mortgage rates reflecting borrower risk, cash-poor households borrow more. These households are more sensitive to flow income and costs, such as rent.